53. FOOD WASTE MONITORING SYSTEM

1. AIM

To monitor daily food preparation and consumption in a hostel dining facility, calculate waste percentages, and raise alerts when waste exceeds safe threshold.

2. PROBLEM STATEMENT

Hostel mess management needs to track daily food waste. Write a Python program that accepts meals prepared and served over multiple days, calculates waste quantity and waste percentage, and prints a warning if waste exceeds 10%.

3. OBJECTIVES

- 1. Input number of days and daily meal statistics.
- 2. Calculate wasted meals = prepared - served.
- 3. Compute waste percentage = (waste / prepared) * 100.
- 4. Warn if waste percentage > 10%.

4. ASSUMPTIONS & REQUIREMENTS

Assumptions:

- Meals served <= Meals prepared.
- Meals prepared is non-negative.

Requirements:

- Python 3.x Interpreter
- Basic arithmetic & decision making

5. ALGORITHM

Step 1: Read number of days.

Step 2: Loop day from 1 to days:

a. Input meals prepared (made) and served.

b. Calculate waste = made - served.

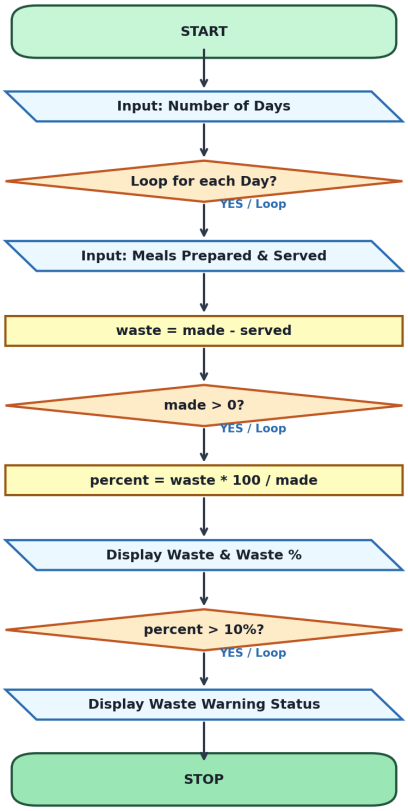
c. If made > 0: percent = waste * 100 / made, else percent = 0.

d. Print waste and round(percent, 2).

e. If percent > 10: print 'High food waste', else print 'Food waste is acceptable'.

Step 3: Stop.

6. PROGRAM FLOWCHART



7. PYTHON PROGRAM (food.py)

```
days = int(input("Enter number of days: "))

for day in range(1, days + 1):
    made = int(input("Meals prepared: "))
    served = int(input("Meals served: "))

    waste = made - served

    if made > 0:
        percent = waste * 100 / made
    else:
        percent = 0

    print("Food waste:", waste)
    print("Waste percentage:", round(percent, 2), "%")

    if percent > 10:
        print("High food waste")
    else:
        print("Food waste is acceptable")
```

8. EXPLANATION OF THE CODE

- **made & served:** Meal counts inputted by mess staff.
- **waste = made - served:** Calculates unused meal count.
- **percent calculation:** Determines ratio of waste relative to prepared food.
- **if percent > 10:** Conditional check triggering high food waste alert.

9. TRACING TABLE

Day	Meals Prepared	Meals Served	Waste	Waste %	Status
Day 1	5	5	0	0.0%	Food waste is acceptable

10. TEST CASES

Test Case	Input	Expected Result	Actual Result
TC 1 (Zero Waste)	Prepared: 5, Served: 5	Food waste: 0 Waste percentage: 0.0% Food waste is acceptable	Food waste: 0 Waste percentage: 0.0% Food waste is acceptable
TC 2 (High Waste)	Prepared: 100, Served: 80	Food waste: 20 Waste percentage: 20.0% High food waste	Food waste: 20 Waste percentage: 20.0% High food waste
TC 3 (Zero Prepared)	Prepared: 0, Served: 0	Food waste: 0 Waste percentage: 0.0% Food waste is acceptable	Food waste: 0 Waste percentage: 0.0% Food waste is acceptable

11. EXPECTED OUTPUT

```
Enter number of days: 2
Meals prepared: 5
Meals served: 5
Food waste: 0
Waste percentage: 0.0 %
Food waste is acceptable
Meals prepared:
```

12. ACTUAL OUTPUT

```
Enter number of days: 2
Meals prepared: 5
Meals served: 5
Food waste: 0
Waste percentage: 0.0 %
Food waste is acceptable
Meals prepared:
```

13. PROGRAM EXECUTION SCREENSHOT

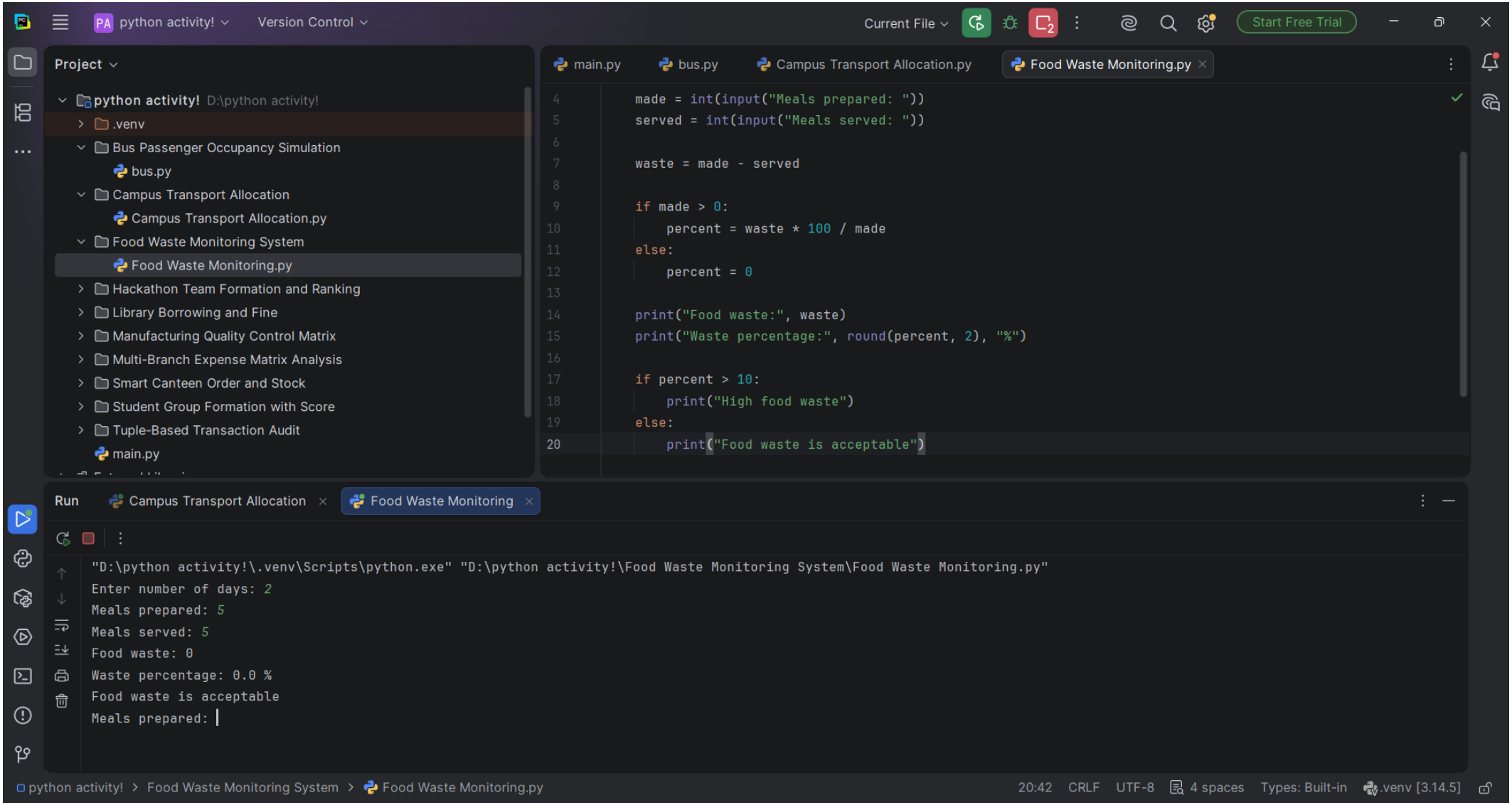
[Screenshot to Be Added]

14. RESULT / CONCLUSION

The Python program food.py was executed successfully. It correctly calculates waste percentage and triggers alerts appropriately.

53 Food Waste Monitoring

Actual Input and Output – PyCharm Run Screenshot



Screenshot includes the program code and the console inputs/outputs.